

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The current prototype does not require user authentication. Future enhancements can include secure login for bank employees, credit officers, and administrators.	Medium
2	Authorization levels	All users can access the loan prediction module. Future versions can provide different access levels where administrators manage datasets and models, while applicants only submit loan details and view predictions.	Medium

3	External interfaces	The system interacts with the trained Machine Learning model (best_loan_model.pkl) and preprocessing objects (preprocessed_loan_data.pkl) through the Flask backend for real-time loan prediction. Future versions can integrate banking APIs and financial data sources.	High
4	Transactions processing	The system accepts applicant information such as income, education, employment status, loan amount, credit history, and property area, preprocesses the data, applies the trained ML model, and returns loan approval or rejection prediction with confidence score.	High
5	Reporting	The system displays loan prediction results including approval/rejection status and confidence percentage. Future versions can include detailed credit analysis reports and dashboards for financial institutions.	Medium
6	Business rules, etc.	User inputs must be validated before prediction. The system uses the same preprocessing steps applied during training, including label encoding and feature scaling, to ensure accurate model predictions.	High
7	Compliance to laws or regulations	The application should follow financial data privacy practices and avoid storing sensitive applicant information without authorization. Future deployments should comply with banking regulations and data protection policies.	Low

8	<i>Other</i>	The application should provide fast and accurate loan eligibility predictions through a simple web interface, helping banks and financial institutions make data-driven decisions.	High
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Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process loan prediction requests quickly after user submission.	Prediction result should be displayed within a few seconds.
2	Scalability	The system should support multiple loan prediction requests and allow future deployment on cloud platforms such as IBM Cloud.	Designed for future scalability and cloud deployment.
3	Security & Data Privacy	The system should validate user inputs and prevent incorrect or malicious data submission. Applicant information should not be stored unnecessarily.	Input validation implemented and no sensitive data stored permanently.

4	Reliability & Availability	The application should provide consistent loan prediction results whenever the system is running.	Reliable execution with stable prediction output during normal operation.
5	Usability & Accessibility	The user interface should be simple, clear, and easy to use for applicants, credit officers, and banking professionals.	Users should complete loan prediction within a short time using the web interface.
6	<i>Other</i>	The application should have modular code structure with separate files for Flask application, prediction logic, models, and templates.	Maintainable Python codebase with reusable components and deployment support.